

Only if Battery already includes Battery Protection: Cut JP1, Bridge JP2, DNP: R1:4, Q1:2, C1, U1

The diagram shows a USB to UART bridge circuit. A Molex PicoBlade 6P connector (J3) provides the USB signals: VBus (pin 6), CC1 (pin 5), CC2 (pin 4), D- (pin 3), D+ (pin 2), and GND (pin 1). The VBus line is connected to a TP20 test point and a 100nF capacitor (C2). The D- and D+ lines are connected to the R22 and R23 resistors (22R), which are in turn connected to the D- and D+ pins of the USBLC6-45C6 chip (U3). The chip's pins 1, 3, 4, and 6 are connected to the R5 (5.1k) and R6 (5.1k) resistors, which are connected to GND. The chip's pins 2 and 5 are connected to the GND pin of the connector. The chip's pins 1 and 3 are connected to the D- and D+ pins of the connector. The chip's pins 4 and 6 are connected to the R5 and R6 resistors, which are connected to GND. The chip's pins 2 and 5 are connected to the GND pin of the connector. The chip's pins 1 and 3 are connected to the D- and D+ pins of the connector. The chip's pins 4 and 6 are connected to the R5 and R6 resistors, which are connected to GND.

CC1 and CC2 pulled low = USB-Device

Keep tracks to U3 short, place C2 close to U3

Place R22:23 close to U10

Interface not used by default

Includes: I2C Pins, UART (or 2 ESP32 GPIOs), JTAG (or 4 free ESP32 GPIOs) and 4 free TCA8418 GPIOs (reroute JP6:9 before use)

Always check I2C Address compatibility when connecting new I2C devices

Extra signals can be connected through TP10:12

The diagram shows a pin header with the following pins and connections:

- TP29: Connected to GND.
- FreeGPIO_B/JTAG_MTD0: Connected to SCL.
- FreeGPIO_7/JTAG_MTD1: Connected to SDA.
- FreeGPIO_4/UART_RX: Connected to TX.
- FreeGPIO_6/JTAG_MTD5: Connected to RX.
- FreeGPIO_3*: Connected to TX.
- FreeGPIO_0*: Connected to RX.
- FreeGPIO_1*: Connected to TX.
- FreeGPIO_5/UART_1X: Connected to TX.
- FreeGPIO_2*: Connected to RX.
- ESP_On: Connected to TX.
- TP10: Connected to TX.
- TP11: Connected to TX.
- TP12: Connected to TX.

JP5 is for testing purposes
Place C9:10 close to U5
Use shielded ferrite core inductors only

+3V3

JP5 can be used during testing to verify if the power supply works as intended (isolation from rest of circuit)

The schematic diagram illustrates the power and data connections for the camera module. The power supply section (U7, U8) and the camera interface section (TP21, TP13) are shown. The power supply section includes two TLV74328PDBV (U7) and TLV74315PDBV (U8) regulators. U7 provides Camera_AVDD (1.8V) and Camera_IOVDD (1.8V). U8 provides Camera_DVDD (1.2V). The camera interface section includes the camera module (TP21) and the camera interface IC (TP13). The camera module has pins for VIN, CT, ON, GND, VOUT, and QOD. The camera interface IC has pins for VIN, CT, ON, GND, VOUT, and QOD. The diagram also shows the connection of the camera module to the camera interface IC via a cable (D2) and the connection of the camera interface IC to the camera module via a cable (R33).

[illegible]

The diagram shows the ESP32-S3-N16R8 module with the following components and connections:

- Power Management:**
 - ESP_Powerlatch:** A red box containing a 10k resistor (R25) and an FDS304P MOSFET (Q3). The gate is connected to +3V3, the drain to the module's +3V3, and the source to GND.
 - ESP_On:** A red box containing an FDS304P MOSFET (Q4). The gate is connected to +3V3, the drain to the module's +3V3, and the source to GND.
 - 3V3_Main_Disable:** A red box containing a 10k resistor (R26) and a SW1 switch. The switch is connected to +3V3, and the resistor is connected to the module's +3V3.
- Resistors:**
 - R24 (100k) connected between the module's EN pin and GND.
 - R38 (2.2k) connected between the module's +3V3 pin and the SW1 switch.
- Capacitors:**
 - C26 (100nF) connected between the module's +3V3 pin and GND.
 - C27 (10μF) connected between the module's +3V3 pin and GND.
- Module Pins and Connections:**
 - GPIOs:** FreeGPIO_4/UART_RX (TP22), FreeGPIO_5/UART_TX (TP25), FreeGPIO_6/JTAG_MTMS (TP26), FreeGPIO_7/JTAG_MTDI (TP27), FreeGPIO_8/JTAG_MTD0 (TP28).
 - Camera:** Camera_VSync (JP10), Camera_HRef (JP11), Camera_D0 (JP24), Camera_D1 (JP35), Camera_D2 (JP36).
 - Keypad:** Keypad_Int (JP26), Keypad_Enable (JP27), Keypad_Res (JP28).
 - Other:** USB_D-/I019, USB_D+/I020, MTCK/I039, MTD0/I040, MTD1/I041, MTMS/I042, TXD0/I043, RXD0/I044, I045, I046, I047, I048.

Use solder jumpers to reroute GPIOs if necessary.

YouTube: @ElectrJonics

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